

GENERIC REFINERY - GULF COAST 100 kBPST Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		AIR COOLER DATASHEET API 661 A-101 - Atm. overhead condenser		DOC. No. CFU-000-ME-DS-005	
				REV / DATE A / 2026-10-02	
				STATUS Issued for review (FEED)	
				PREPARED CLAUDE CODE CHK: - APP: -	
				SHEET 1	
PERFORMANCE DATA					
1	Service	Atm. overhead condenser	2	Heat exchanged	33.75 MW
3	Fluid	OH vapour + steam (sour)	4	Total flow	229.2 t/h
5	Phase	Condensing	6	Inlet / outlet temperature	134 / 60 C
7	Allowable pressure drop	0.35 bar	8	Fouling resistance	0.00035 m2K/W
9	Design air temperature in / out	35 / 55 C	10	Air flow (total)	1677 kg/s
11	Elevation	5 m ASL			
CONSTRUCTION - TUBE BUNDLE					
12	Type / draft	Forced draft, pipe-rack mounted	13	Bays (data) / bays required @ 6 rows	3 / 4
14	Bay size	6.0 m W x 12.0 m L	15	Bundles per bay	2
16	Tube rows (calc. for data bays)	7	17	Bare surface (total)	1667 m2
18	Extended surface (approx. x21)	35,012 m2	19	Tubes: OD / thk / length	25.4 mm / 2.77 mm / 12.0 m
20	Tube material	A214 ERW CS + inlet ferrules	21	Fins	Al 1100, L-footed / extruded, 57.2 mm OD
22	Header type / plugs	Plug-type box / shoulder plugs, SS	23	Design pressure / temperature	9.7 barg / 165 C
24	Test pressure	12.6 barg	25	Corrosion allowance	3 mm headers / 0 tubes
26	Nozzle rating	150# RF	27	Code	ASME VIII-1 / API 661
MECHANICAL EQUIPMENT					
28	Fans per bay / total	2 / 6	29	Fan diameter	4.5 m (>= 40 % coverage)
30	Blades / pitch	FRP or Al; manual pitch, 1 auto-variable pitch fan p-	31	Motor rated / absorbed per fan	55 / 45.0 kW
32	Motor	480 V, 3 ph, 60 Hz, TEFC, Ex nA / Class I Div 2	33	Drive	HTD belt (<= 45 kW) / right-angle gear (> 45 kW)
34	Speed control	VFD on 1 fan per bay (TBD)	35	Vibration switches	Yes, each fan
36	Louvres / winterisation	None (Gulf Coast); A-101 inlet louvres optional	37	Noise	<= 85 dB(A) at 1 m below bundle
38	Weight empty / operating (est.)	118.7 / 123.9 t			
NOTES					
1. API 661 datasheet layout. Thermal rating by vendor.					

GENERIC REFINERY - GULF COAST 100 kBPST Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		AIR COOLER DATASHEET API 661 A-103 - Kero cooler		DOC. No. CFU-000-ME-DS-005	
				REV / DATE A / 2026-10-02	
				STATUS Issued for review (FEED)	
				PREPARED CLAUDE CODE CHK: - APP: -	
				SHEET 2	
PERFORMANCE DATA					
1	Service	Kero cooler	2	Heat exchanged	1.27 MW
3	Fluid	Kero	4	Total flow	68.6 t/h
5	Phase	Liquid	6	Inlet / outlet temperature	76 / 45 C
7	Allowable pressure drop	0.7 bar	8	Fouling resistance	0.00035 m2K/W
9	Design air temperature in / out	35 / 40 C	10	Air flow (total)	252 kg/s
11	Elevation	5 m ASL			
CONSTRUCTION - TUBE BUNDLE					
12	Type / draft	Forced draft, pipe-rack mounted	13	Bays (data) / bays required @ 6 rows	1 / 1
14	Bay size	6.0 m W x 12.0 m L	15	Bundles per bay	2
16	Tube rows (calc. for data bays)	3	17	Bare surface (total)	184 m2
18	Extended surface (approx. x21)	3,857 m2	19	Tubes: OD / thk / length	25.4 mm / 2.77 mm / 12.0 m
20	Tube material	A214 ERW CS	21	Fins	Al 1100, L-footed / extruded, 57.2 mm OD
22	Header type / plugs	Plug-type box / shoulder plugs, SS	23	Design pressure / temperature	9.7 barg / 105 C
24	Test pressure	12.6 barg	25	Corrosion allowance	3 mm headers / 0 tubes
26	Nozzle rating	150# RF	27	Code	ASME VIII-1 / API 661
MECHANICAL EQUIPMENT					
28	Fans per bay / total	2 / 2	29	Fan diameter	4.5 m (>= 40 % coverage)
30	Blades / pitch	FRP or Al; manual pitch, 1 auto-variable pitch fan p-	31	Motor rated / absorbed per fan	11 / 7.5 kW
32	Motor	480 V, 3 ph, 60 Hz, TEFC, Ex nA / Class I Div 2	33	Drive	HTD belt (<= 45 kW) / right-angle gear (> 45 kW)
34	Speed control	VFD on 1 fan per bay (TBD)	35	Vibration switches	Yes, each fan
36	Louvres / winterisation	None (Gulf Coast); A-101 inlet louvres optional	37	Noise	<= 85 dB(A) at 1 m below bundle
38	Weight empty / operating (est.)	23.8 / 24.4 t			
NOTES					
1. API 661 datasheet layout. Thermal rating by vendor.					

GENERIC REFINERY - GULF COAST 100 kBPST Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		AIR COOLER DATASHEET API 661 A-104 - Diesel cooler		DOC. No.	CFU-000-ME-DS-005
				REV / DATE	A / 2026-10-02
				STATUS	Issued for review (FEED)
				PREPARED	CLAUDE CODE CHK: - APP: -
				SHEET	3

PERFORMANCE DATA					
1	Service	Diesel cooler	2	Heat exchanged	3.72 MW
3	Fluid	Diesel	4	Total flow	81.6 t/h
5	Phase	Liquid	6	Inlet / outlet temperature	128 / 55 C
7	Allowable pressure drop	0.7 bar	8	Fouling resistance	0.00035 m2K/W
9	Design air temperature in / out	35 / 50 C	10	Air flow (total)	247 kg/s
11	Elevation	5 m ASL			

CONSTRUCTION - TUBE BUNDLE					
12	Type / draft	Forced draft, pipe-rack mounted	13	Bays (data) / bays required @ 6 rows	1 / 1
14	Bay size	6.0 m W x 12.0 m L	15	Bundles per bay	2
16	Tube rows (calc. for data bays)	4	17	Bare surface (total)	302 m2
18	Extended surface (approx. x21)	6,352 m2	19	Tubes: OD / thk / length	25.4 mm / 2.77 mm / 12.0 m
20	Tube material	A214 ERW CS	21	Fins	Al 1100, L-footed / extruded, 57.2 mm OD
22	Header type / plugs	Plug-type box / shoulder plugs, SS	23	Design pressure / temperature	9.7 barg / 160 C
24	Test pressure	12.6 barg	25	Corrosion allowance	3 mm headers / 0 tubes
26	Nozzle rating	150# RF	27	Code	ASME VIII-1 / API 661

MECHANICAL EQUIPMENT					
28	Fans per bay / total	2 / 2	29	Fan diameter	4.5 m (>= 40 % coverage)
30	Blades / pitch	FRP or Al; manual pitch, 1 auto-variable pitch fan p-	31	Motor rated / absorbed per fan	18.5 / 14.9 kW
32	Motor	480 V, 3 ph, 60 Hz, TEFC, Ex nA / Class I Div 2	33	Drive	HTD belt (<= 45 kW) / right-angle gear (> 45 kW)
34	Speed control	VFD on 1 fan per bay (TBD)	35	Vibration switches	Yes, each fan
36	Louvres / winterisation	None (Gulf Coast); A-101 inlet louvres optional	37	Noise	<= 85 dB(A) at 1 m below bundle
38	Weight empty / operating (est.)	28.8 / 29.8 t			

NOTES					
1. API 661 datasheet layout. Thermal rating by vendor.					

GENERIC REFINERY - GULF COAST 100 kBPSD Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		AIR COOLER DATASHEET API 661 A-105 - Ago cooler		DOC. No. CFU-000-ME-DS-005	
				REV / DATE A / 2026-10-02	
				STATUS Issued for review (FEED)	
				PREPARED CLAUDE CODE CHK: - APP: -	
				SHEET 4	
PERFORMANCE DATA					
1	Service	Ago cooler	2	Heat exchanged	6.06 MW
3	Fluid	Ago	4	Total flow	45.8 t/h
5	Phase	Liquid	6	Inlet / outlet temperature	254 / 60 C
7	Allowable pressure drop	0.7 bar	8	Fouling resistance	0.00035 m2K/W
9	Design air temperature in / out	35 / 55 C	10	Air flow (total)	301 kg/s
11	Elevation	5 m ASL			
CONSTRUCTION - TUBE BUNDLE					
12	Type / draft	Induced draft, pipe-rack mounted	13	Bays (data) / bays required @ 6 rows	1 / 1
14	Bay size	6.0 m W x 12.0 m L	15	Bundles per bay	2
16	Tube rows (calc. for data bays)	4	17	Bare surface (total)	287 m2
18	Extended surface (approx. x21)	6,019 m2	19	Tubes: OD / thk / length	25.4 mm / 2.77 mm / 12.0 m
20	Tube material	A214 ERW CS	21	Fins	Al 1100, embedded (G-fin) 57.2 mm OD, 433 fins/m
22	Header type / plugs	Plug-type box / shoulder plugs, SS	23	Design pressure / temperature	9.7 barg / 285 C
24	Test pressure	12.6 barg	25	Corrosion allowance	3 mm headers / 0 tubes
26	Nozzle rating	150# RF	27	Code	ASME VIII-1 / API 661
MECHANICAL EQUIPMENT					
28	Fans per bay / total	2 / 2	29	Fan diameter	4.5 m (>= 40 % coverage)
30	Blades / pitch	FRP or Al; manual pitch, 1 auto-variable pitch fan p-	31	Motor rated / absorbed per fan	30 / 24.2 kW
32	Motor	480 V, 3 ph, 60 Hz, TEFC, Ex nA / Class I Div 2	33	Drive	HTD belt (<= 45 kW) / right-angle gear (> 45 kW)
34	Speed control	VFD on 1 fan per bay (TBD)	35	Vibration switches	Yes, each fan
36	Louvres / winterisation	None (Gulf Coast); A-101 inlet louvres optional	37	Noise	<= 85 dB(A) at 1 m below bundle
38	Weight empty / operating (est.)	28.2 / 29.1 t			
NOTES					
1. API 661 datasheet layout. Thermal rating by vendor.					

GENERIC REFINERY - GULF COAST 100 kBPST Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		AIR COOLER DATASHEET API 661 A-106 - Stabiliser overhead condenser		DOC. No. CFU-000-ME-DS-005	
				REV / DATE A / 2026-10-02	
				STATUS Issued for review (FEED)	
				PREPARED CLAUDE CODE CHK: - APP: -	
				SHEET 5	
PERFORMANCE DATA					
1	Service	Stabiliser overhead condenser	2	Heat exchanged	2.14 MW
3	Fluid	Stabiliser	4	Total flow	18.7 t/h
5	Phase	Condensing	6	Inlet / outlet temperature	79 / 45 C
7	Allowable pressure drop	0.35 bar	8	Fouling resistance	0.00035 m2K/W
9	Design air temperature in / out	35 / 40 C	10	Air flow (total)	425 kg/s
11	Elevation	5 m ASL			
CONSTRUCTION - TUBE BUNDLE					
12	Type / draft	Forced draft, pipe-rack mounted	13	Bays (data) / bays required @ 6 rows	1 / 1
14	Bay size	6.0 m W x 12.0 m L	15	Bundles per bay	2
16	Tube rows (calc. for data bays)	3	17	Bare surface (total)	214 m2
18	Extended surface (approx. x21)	4,492 m2	19	Tubes: OD / thk / length	25.4 mm / 2.77 mm / 12.0 m
20	Tube material	A214 ERW CS	21	Fins	Al 1100, L-footed / extruded, 57.2 mm OD
22	Header type / plugs	Plug-type box / shoulder plugs, SS	23	Design pressure / temperature	16.7 barg / 110 C
24	Test pressure	21.7 barg	25	Corrosion allowance	3 mm headers / 0 tubes
26	Nozzle rating	150# RF	27	Code	ASME VIII-1 / API 661
MECHANICAL EQUIPMENT					
28	Fans per bay / total	2 / 2	29	Fan diameter	4.5 m (>= 40 % coverage)
30	Blades / pitch	FRP or Al; manual pitch, 1 auto-variable pitch fan p-	31	Motor rated / absorbed per fan	11 / 8.6 kW
32	Motor	480 V, 3 ph, 60 Hz, TEFC, Ex nA / Class I Div 2	33	Drive	HTD belt (<= 45 kW) / right-angle gear (> 45 kW)
34	Speed control	VFD on 1 fan per bay (TBD)	35	Vibration switches	Yes, each fan
36	Louvres / winterisation	None (Gulf Coast); A-101 inlet louvres optional	37	Noise	<= 85 dB(A) at 1 m below bundle
38	Weight empty / operating (est.)	25.1 / 25.8 t			
NOTES					
1. API 661 datasheet layout. Thermal rating by vendor.					

GENERIC REFINERY - GULF COAST 100 kBPSP Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		AIR COOLER DATASHEET API 661 A-107 - Splitter overhead condenser		DOC. No. CFU-000-ME-DS-005	
				REV / DATE A / 2026-10-02	
				STATUS Issued for review (FEED)	
				PREPARED CLAUDE CODE CHK: - APP: -	
				SHEET 6	
PERFORMANCE DATA					
1	Service	Splitter overhead condenser	2	Heat exchanged	9.63 MW
3	Fluid	Splitter	4	Total flow	85.1 t/h
5	Phase	Condensing	6	Inlet / outlet temperature	84 / 50 C
7	Allowable pressure drop	0.35 bar	8	Fouling resistance	0.00035 m2K/W
9	Design air temperature in / out	35 / 45 C	10	Air flow (total)	957 kg/s
11	Elevation	5 m ASL			
CONSTRUCTION - TUBE BUNDLE					
12	Type / draft	Forced draft, pipe-rack mounted	13	Bays (data) / bays required @ 6 rows	2 / 2
14	Bay size	6.0 m W x 12.0 m L	15	Bundles per bay	2
16	Tube rows (calc. for data bays)	6	17	Bare surface (total)	881 m2
18	Extended surface (approx. x21)	18,503 m2	19	Tubes: OD / thk / length	25.4 mm / 2.77 mm / 12.0 m
20	Tube material	A214 ERW CS	21	Fins	Al 1100, L-footed / extruded, 57.2 mm OD
22	Header type / plugs	Plug-type box / shoulder plugs, SS	23	Design pressure / temperature	9.7 barg / 115 C
24	Test pressure	12.6 barg	25	Corrosion allowance	3 mm headers / 0 tubes
26	Nozzle rating	150# RF	27	Code	ASME VIII-1 / API 661
MECHANICAL EQUIPMENT					
28	Fans per bay / total	2 / 4	29	Fan diameter	4.5 m (>= 40 % coverage)
30	Blades / pitch	FRP or Al; manual pitch, 1 auto-variable pitch fan p-	31	Motor rated / absorbed per fan	30 / 19.3 kW
32	Motor	480 V, 3 ph, 60 Hz, TEFC, Ex nA / Class I Div 2	33	Drive	HTD belt (<= 45 kW) / right-angle gear (> 45 kW)
34	Speed control	VFD on 1 fan per bay (TBD)	35	Vibration switches	Yes, each fan
36	Louvres / winterisation	None (Gulf Coast); A-101 inlet louvres optional	37	Noise	<= 85 dB(A) at 1 m below bundle
38	Weight empty / operating (est.)	69.4 / 72.1 t			
NOTES					
1. API 661 datasheet layout. Thermal rating by vendor.					

GENERIC REFINERY - GULF COAST 100 kBPSP Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		AIR COOLER DATASHEET API 661 A-108 - Heavy naphtha product cooler		DOC. No. CFU-000-ME-DS-005	
				REV / DATE A / 2026-10-02	
				STATUS Issued for review (FEED)	
				PREPARED CLAUDE CODE CHK: - APP: -	
				SHEET 7	
PERFORMANCE DATA					
1	Service	Heavy naphtha product cooler	2	Heat exchanged	5.81 MW
3	Fluid	Heavy	4	Total flow	80.8 t/h
5	Phase	Liquid	6	Inlet / outlet temperature	152 / 45 C
7	Allowable pressure drop	0.7 bar	8	Fouling resistance	0.00035 m2K/W
9	Design air temperature in / out	35 / 40 C	10	Air flow (total)	1155 kg/s
11	Elevation	5 m ASL			
CONSTRUCTION - TUBE BUNDLE					
12	Type / draft	Induced draft, pipe-rack mounted	13	Bays (data) / bays required @ 6 rows	1 / 1
14	Bay size	6.0 m W x 12.0 m L	15	Bundles per bay	2
16	Tube rows (calc. for data bays)	5	17	Bare surface (total)	401 m2
18	Extended surface (approx. x21)	8,431 m2	19	Tubes: OD / thk / length	25.4 mm / 2.77 mm / 12.0 m
20	Tube material	A214 ERW CS	21	Fins	Al 1100, L-footed / extruded, 57.2 mm OD
22	Header type / plugs	Plug-type box / shoulder plugs, SS	23	Design pressure / temperature	9.7 barg / 185 C
24	Test pressure	12.6 barg	25	Corrosion allowance	3 mm headers / 0 tubes
26	Nozzle rating	150# RF	27	Code	ASME VIII-1 / API 661
MECHANICAL EQUIPMENT					
28	Fans per bay / total	2 / 2	29	Fan diameter	4.5 m (>= 40 % coverage)
30	Blades / pitch	FRP or Al; manual pitch, 1 auto-variable pitch fan p~	31	Motor rated / absorbed per fan	30 / 23.2 kW
32	Motor	480 V, 3 ph, 60 Hz, TEFC, Ex nA / Class I Div 2	33	Drive	HTD belt (<= 45 kW) / right-angle gear (> 45 kW)
34	Speed control	VFD on 1 fan per bay (TBD)	35	Vibration switches	Yes, each fan
36	Louvres / winterisation	None (Gulf Coast); A-101 inlet louvres optional	37	Noise	<= 85 dB(A) at 1 m below bundle
38	Weight empty / operating (est.)	33.0 / 34.3 t			
NOTES					
1. API 661 datasheet layout. Thermal rating by vendor.					

GENERIC REFINERY - GULF COAST 100 kBPST Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: VDU		AIR COOLER DATASHEET API 661 A-201 - Lvgo cooler		DOC. No.	CFU-000-ME-DS-005
				REV / DATE	A / 2026-10-02
				STATUS	Issued for review (FEED)
				PREPARED	CLAUDE CODE CHK: - APP: -
				SHEET	8
PERFORMANCE DATA					
1	Service	Lvgo cooler	2	Heat exchanged	3.10 MW
3	Fluid	Lvgo	4	Total flow	132.9 t/h
5	Phase	Liquid	6	Inlet / outlet temperature	95 / 55 C
7	Allowable pressure drop	0.7 bar	8	Fouling resistance	0.00035 m2K/W
9	Design air temperature in / out	35 / 50 C	10	Air flow (total)	205 kg/s
11	Elevation	5 m ASL			
CONSTRUCTION - TUBE BUNDLE					
12	Type / draft	Forced draft, pipe-rack mounted	13	Bays (data) / bays required @ 6 rows	1 / 1
14	Bay size	6.0 m W x 12.0 m L	15	Bundles per bay	2
16	Tube rows (calc. for data bays)	5	17	Bare surface (total)	397 m2
18	Extended surface (approx. x21)	8,328 m2	19	Tubes: OD / thk / length	25.4 mm / 2.77 mm / 12.0 m
20	Tube material	A214 ERW CS	21	Fins	Al 1100, L-footed / extruded, 57.2 mm OD
22	Header type / plugs	Plug-type box / shoulder plugs, SS	23	Design pressure / temperature	9.7 barg / 125 C
24	Test pressure	12.6 barg	25	Corrosion allowance	3 mm headers / 0 tubes
26	Nozzle rating	150# RF	27	Code	ASME VIII-1 / API 661
MECHANICAL EQUIPMENT					
28	Fans per bay / total	2 / 2	29	Fan diameter	4.5 m (>= 40 % coverage)
30	Blades / pitch	FRP or Al; manual pitch, 1 auto-variable pitch fan p~	31	Motor rated / absorbed per fan	15 / 12.4 kW
32	Motor	480 V, 3 ph, 60 Hz, TEFC, Ex nA / Class I Div 2	33	Drive	HTD belt (<= 45 kW) / right-angle gear (> 45 kW)
34	Speed control	VFD on 1 fan per bay (TBD)	35	Vibration switches	Yes, each fan
36	Louvres / winterisation	None (Gulf Coast); A-101 inlet louvres optional	37	Noise	<= 85 dB(A) at 1 m below bundle
38	Weight empty / operating (est.)	32.8 / 34.1 t			
NOTES					
1. API 661 datasheet layout. Thermal rating by vendor.					

GENERIC REFINERY - GULF COAST 100 kBPST Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: VDU		AIR COOLER DATASHEET API 661 A-202 - Hvgo Product cooler		DOC. No. CFU-000-ME-DS-005	
				REV / DATE A / 2026-10-02	
				STATUS Issued for review (FEED)	
				PREPARED CLAUDE CODE CHK: - APP: -	
				SHEET 9	
PERFORMANCE DATA					
1	Service	Hvgo Product cooler	2	Heat exchanged	6.64 MW
3	Fluid	Hvgo	4	Total flow	90.4 t/h
5	Phase	Liquid	6	Inlet / outlet temperature	205 / 90 C
7	Allowable pressure drop	0.7 bar	8	Fouling resistance	0.00035 m2K/W
9	Design air temperature in / out	35 / 60 C	10	Air flow (total)	264 kg/s
11	Elevation	5 m ASL			
CONSTRUCTION - TUBE BUNDLE					
12	Type / draft	Induced draft, pipe-rack mounted	13	Bays (data) / bays required @ 6 rows	1 / 1
14	Bay size	6.0 m W x 12.0 m L	15	Bundles per bay	2
16	Tube rows (calc. for data bays)	5	17	Bare surface (total)	361 m2
18	Extended surface (approx. x21)	7,586 m2	19	Tubes: OD / thk / length	25.4 mm / 2.77 mm / 12.0 m
20	Tube material	A214 ERW CS	21	Fins	Al 1100, embedded (G-fin) 57.2 mm OD, 433 fins/m
22	Header type / plugs	Plug-type box / shoulder plugs, SS	23	Design pressure / temperature	9.7 barg / 235 C
24	Test pressure	12.6 barg	25	Corrosion allowance	3 mm headers / 0 tubes
26	Nozzle rating	150# RF	27	Code	ASME VIII-1 / API 661
MECHANICAL EQUIPMENT					
28	Fans per bay / total	2 / 2	29	Fan diameter	4.5 m (>= 40 % coverage)
30	Blades / pitch	FRP or Al; manual pitch, 1 auto-variable pitch fan p-	31	Motor rated / absorbed per fan	37 / 26.6 kW
32	Motor	480 V, 3 ph, 60 Hz, TEFC, Ex nA / Class I Div 2	33	Drive	HTD belt (<= 45 kW) / right-angle gear (> 45 kW)
34	Speed control	VFD on 1 fan per bay (TBD)	35	Vibration switches	Yes, each fan
36	Louvres / winterisation	None (Gulf Coast); A-101 inlet louvres optional	37	Noise	<= 85 dB(A) at 1 m below bundle
38	Weight empty / operating (est.)	31.3 / 32.5 t			
NOTES					
1. API 661 datasheet layout. Thermal rating by vendor.					